NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND
MIL-PRF-82791(OS)
5 February 1990
SUPERSEDING
WS 20614
24 May 1988

PERFORMANCE SPECIFICATION

RUBBER, SILICONE, RTV

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification defines one type of room temperature vulcanized (RTV) silicone rubber referred to herein as the rubber, see (6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-S-5059 Steel, Corrosion-Resistant (18-8) Plate, Sheet and Strip

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, Maryland 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1340

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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STANDARDS

MILITARY

MIL-STD-129 Marking for shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications standards and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues o documents not listed in the DODISS are the issues of the documents cited in the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 412 Rubber Properties in Tension (DoD adopted)

ASTM D 624 Rubber Property - Tear Resistance (DoD adopted)

ASTM D 746 Brittleness Temperature of Plastics and Elastomers by Impact (DoD adopted)

ASTM D 792 Specific Gravity (Relative Density) and Density of Plastics by Displacement (DoD adopted)

ASTM D 1084 Viscosity of Adhesives

ASTM D 2240 Rubber Property - Durometer Hardness (DoD adopted)

ASTM D 3182 Rubber - Materials, Equipment and Procedures for Mixing Standard Compounds and Preparing Standard Vulcanized Sheets

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

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AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1120, Chicago, IL 60606-5945.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **First article.** When specified (see 6.2), a sample of the rubber shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 **Material.** The rubber furnished in accordance with this specification shall be a two-part compound consisting of a silicone rubber-base material and a curing agent. The rubber shall cure at room temperature after addition of the curing agent (catalyst). The base material shall be of the dimethyl silicone type.

3.3 **Chemical and physical properties.** Chemical and physical properties of the material shall conform to the requirements specified in table I.

TABLE I. Chemical and physical properties

Property	Value 1/
Viscosity, poise, maximum	
Base material	500 to 1600
Catalyzed, 0 hours 2/	1000 maximum
Catalyzed, 1.5 hours 2/	2000 maximum
Specific gravity, cured material	1.07 to 1.17

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1/ Determined at 77 ± 2 degrees Fahrenheit (°F) and 50 ± 5 percent relative humidity.
2/ Catalyzed using the amount of curing agent recommended by the manufacturer.

3.4 **Mechanical properties.** The mechanical properties of the cured rubber be as specified in table II.

TABLE II. Cured rubber mechanical properties.

Property	Value
Hardness, Shore A, minimum	30
Tensile strength, psi, minimum	600
Elongation, percent, minimum	250
Tear resistance, pounds per inch, minimum	80

3.5 **Brittleness temperature.** The brittleness temperature of the rubber shall be -65°F or lower. Brittleness temperature is the lowest temperature at which no failures occur among five specimens tested. At the brittleness temperature, there shall be no cracking, fissures, rupturing, or crazing visible to the unaided eye when tested as specified herein.

3.6 **Shelf life.** The rubber shall meet the requirements of this specification, before mixing, for a period of 6 months from the date of the quality conformance inspection (see 4.4) when packaged as specified in section 5 and stored under the conditions marked on the container. The required shelf life is based on storage in original containers at a temperature not greater than 90°F.

3.6.1 **Shelf life extension.** The shelf life extension of the rubber before mixing, may be extended for 3 months on the basis of successful completion of acceptance tests, as specified herein, for viscosity, hardness, tensile strength and elongation.

3.7 **Workmanship.** Workmanship shall be such that the rubber before mixing, is uniform in quality and color and free from impurities and lumps.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspections.** Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the

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Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein shall be classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the requirements specified in 4.4. Failure to meet all of the requirements specified herein shall be cause for rejection of the first article sample.

4.4 Quality conformance inspection. Quality conformance inspections shall consist of the tests specified in table III:

TABLE III. Quality conformance inspection.

Tests	Requirement	Test method
Viscosity test	3.3	4.5.1
Shore A hardness test	3.4	4.5.2
Tensile strength test	3.4	4.5.3
Elongation test	3.4	4.5.4
Tear resistance test	3.4	4.5.5
Brittleness test	3.5	4.5.6
Specific gravity test	3.3	4.5.7

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4.4.1 **Sampling for inspection.**

4.4.1.1 **Lot.** A lot shall consist of that quantity of each of the two components of the sealant material which is manufactured at one time, in one batch (see 6.4) and inspected at one time.

4.4.1.2 **Selection of samples.** Samples shall be selected at random from each lot or batch. The samples shall be of sufficient size to enable performance of all tests specified herein.

4.4.1.3 **Specimen preparation.** Test specimens shall be molded in conformance with ASTM D 3811 molds for the following tests:

a. Specific gravity.

b. Hardness, Shore A.

c. Tensile strength.

d. Elongation.

e. Tear resistance.

4.4.1.4 **Molding of specimens.** Thoroughly mix the silicone rubber base compound and curing agent in proportions of 10 parts base to 1 part curing agent. Pour the mixed compound into the mold. Close the mold and cure for 24 hours, minimum, at 77 ± 5°F.

4.5 **Test methods.**

4.5.1 **Viscosity test.** The viscosity measurement shall be determined in accordance with ASTM D 1084, method B. A Brookfield Model HAF or equivalent, shall be used with a No. 6 spindle at 10 revolutions per minute (RPM). The reading shall be made at 1 minute from the time the run is started. Viscosities shall be recorded in poise.

4.5.1.1 **Base material test.** The base material shall be stirred for 5 ± 1 minutes, degassed under vacuum, and the viscosity measured immediately.

4.5.1.2 **Catalyzed material test.** The base and curing agent materials shall be mixed for 5 ± 1 minutes and degassed under vacuum. The viscosity shall be measured immediately and at 90 ± 10 minutes.

4.5.2 **Shore A hardness test.** Shore A hardness of the cured rubber shall be determined in accordance with ASTM D 2240.

4.5.3 **Tensile strength test.** The tensile strength of the cured rubber shall be determined in accordance with ASTM D 412, using Die C.

4.5.4 **Elongation test.** Elongation of the cured rubber shall be determined in accordance with ASTM D 412, using Die C.

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4.5.5 Tear resistance test. The tear resistance of the cured rubber shall be determined in accordance with ASTM D 624, using Die B.

4.5.6 Brittleness temperature test. The cured rubber shall be tested in accordance with ASTM D 746 at a temperature of -65°F, using Inspection and Acceptance Procedure B to verify that the brittleness temperature conforms to 3.5. Failure of any of the five specimens tested to meet the requirements of this specification shall be cause for rejection of the lot represented.

4.5.7 Specific gravity test. The specific gravity of the cured sealant shall be determined in accordance with ASTM D 792, method A.

4.6 Inspection of packaging. The silicone rubber containers and container markings shall be visually examined for compliance with section 5.

5. PACKAGING

5.1 Preservation and packaging. Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C as specified herein. Each component of the rubber shall be packaged separately in sealed containers.

5.2 Level C. The rubber shall be packaged to afford adequate protection against loss, contamination, deterioration, and damage during shipment from the supply source to the first receiving activity and during storage under the shelf life period and conditions specified in 3.6. Containers in the same shipment shall be of the same size. The packaging shall conform to UFC 6000, National Motor Freight Classification, 49 CFR 171-178, or to other carrier rules and regulations as applicable to the mode of transportation.

5.3 Marking. Unless otherwise specified in the contract or purchase order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.3.1 Special marking. In addition to the marking requirements of 5.3, each container marking shall include the following information:

a. Title, number and date of this specification.

b. Manufacturer's name and address.

c. Date of manufacture.

d. Lot number and date of quality conformance inspection.

f. Handling/mixing instructions.

g. Shelf life and inspection/test date for extension (see 3.6.1). In addition, the marking shall include the following:

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"THE SHELF LIFE OF AN INDIVIDUAL LOT MAY BE EXTENDED FOR AN ADDITIONAL 3 MONTH PERIOD PROVIDED THE RUBBER, UPON RETEST, CONFORMS TO THE CHEMICAL AND PHYSICAL PROPERTIES REQUIREMENTS SPECIFIED IN MIL-R-82791."

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The rubber specified herein is intended for use as a component on solid rocket motors.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

c. Whether first article inspection is required (see 3.1).

6.3 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors on the materia to be inspected. The material should be a first article sample and should consist of a quantity of the material sufficient to perform all of the inspections specified in 4.3. The first article sample should be packaged and marked as specified in section 5. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definition.

6.4.1 Batch. A batch is that quantity of rubber which has been manufactured by the same unit chemical process and subjected to the same physical mixing operation intended to make the final product substantially uniform.

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6.5 Subject term (key word) listing.

Rocket Motor, Dual Thrust, Mk 104
Standard Missile

Preparing Activity:
NAVY-OS
(Project 1340-N798)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-R-82791	2. DOCUMENT DATE (YYMMDD) 5 February 1990

3. DOCUMENT TITLE
RUBBER SILICONE, RTV

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

b. NAME (Last, First, Middle Initial)	c. ORGANIZATION	
d. ADDRESS (Include Zip Code)	e. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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